

Impact of complete colonoscopy after ischaemic colitis onset on colorectal neoplasm detection in patients without suspected coexistence of colorectal cancer on computed tomography: a retrospective analysis

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ABSTRACT

Objective Ischaemic colitis is the most prevalent form of ischaemic enteritis and represents a major cause of acute lower gastrointestinal bleeding. Although the American College of Gastroenterology's clinical guidelines recommend colonoscopy after ischaemic colitis to screen for colorectal cancer, the actual detection rate of neoplastic lesions in patients without suspected malignancies on CT remains unclear. This study aimed to assess the efficacy of colonoscopy in detecting colorectal neoplasms after the resolution of ischaemic colitis.

Methods This retrospective, single-centre, observational study included patients diagnosed with ischaemic colitis at the Iseaki Municipal Hospital in Japan between 2014 and 2023. Patients with CT-confirmed ischaemic colitis without a suspicion of colorectal cancer were eligible. Clinical data, colonoscopic findings and histopathological results were extracted from medical records. Comparative analyses were conducted between patients who underwent complete colonoscopy and those who did not.

Results Among the 418 patients diagnosed with ischaemic colitis, 396 underwent CT imaging, and 116 underwent subsequent complete colonoscopy. Colorectal polyps were identified in 34.5% (40/116) of the patients, with 75 lesions predominantly located in the right-sided colon. Invasive colorectal carcinoma was detected in 3.4% (4/116) of the patients, along with an additional case of intramucosal carcinoma. Notably, one invasive adenocarcinoma was located proximal to the site of the ischaemic injury. Between the complete colonoscopy and incomplete/no colonoscopy groups, the patients in the incomplete/no colonoscopy group were significantly older, had a higher proportion of poor performance status and were more likely to have used saline laxatives.

Conclusion Colonoscopy after ischaemic colitis revealed a non-negligible prevalence of colorectal neoplasms even in the absence of CT findings suggestive of malignancies. These results underscore the importance of colonoscopy after recovery, particularly in patients without a poor

WHAT IS ALREADY KNOWN ON THIS TOPIC

⇒ Ischaemic colitis (IC) is a common cause of acute lower gastrointestinal bleeding and is rarely associated with colorectal cancer (CRC). Current guidelines recommend colonoscopy after IC recovery to screen for CRC, but the utility of such screening in patients without CT suspicion of malignancy has not been well defined.

WHAT THIS STUDY ADDS

⇒ This study demonstrates that complete colonoscopy after IC reveals a substantial prevalence of invasive colorectal carcinoma in 3.4% of patients, even when CT does not indicate malignancy. A significant proportion of lesions was located in the right-sided colon and was not necessarily related to the ischaemic site.

HOW THIS STUDY MIGHT AFFECT RESEARCH, PRACTICE OR POLICY

⇒ Our findings support the routine implementation of complete colonoscopy following IC recovery in patients with adequate performance status, regardless of CT findings. This may enhance early detection of asymptomatic colorectal neoplasms and inform future screening strategies after IC.

performance status. Further prospective, multicentre studies are warranted to validate these findings and optimise postischaemic colitis management strategies.

INTRODUCTION

Ischaemic colitis (IC), a local hypoperfusion and reperfusion disorder, is the most frequent form of ischaemic enteritis and a common cause of acute lower gastrointestinal

bleeding.^{1–4} Clinically, IC can be classified as gangrenous or non-gangrenous, and most cases are non-gangrenous and self-limiting, with few cases leading to serious outcomes.⁵ IC symptoms generally involve sudden abdominal pain, diarrhoea and bloody stools.⁶ Furthermore, the development of IC is associated with a series of diseases and risk factors. Any condition or factor that decreases blood flow to the colon can cause IC. Therefore, risk factors include a history of vascular surgery (specifically aortic or coronary artery bypass surgery) as well as hypertension, diabetes mellitus, hyperlipidaemia and microvascular diseases such as systemic lupus erythematosus.^{7–9} Other risk factors include constipation-inducing drugs, immunomodulators and illicit drugs.¹⁰

Although rare, IC can be triggered by stenosis due to colorectal cancer (CRC), and the coexistence of both conditions on CT is reported in approximately 5% of the patients with IC.¹¹ According to the American College of Gastroenterology (ACG) clinical guidelines, CRC screening via colonoscopy should be considered several weeks after recovery from ischaemic injury, despite the infrequent association between IC and distal malignancies.¹² In addition, patients with IC may have a high risk of CRC due to overlapping risk factors for CRC, such as older age, diabetes mellitus and smoking. However, there are no reports examining the prevalence of colorectal neoplasms detected on colonoscopy in patients with IC who showed no suspicion of CRC on CT. This study aimed to investigate the impact of complete colonoscopy after IC onset on the detection of colorectal neoplasms in patients without suspected coexisting CRC on CT.

METHODS

Study population

This retrospective observational study included patients diagnosed with IC between January 2014 and December 2023 at Isesaki Municipal Hospital, a community-based general hospital in Gunma, Japan. The Strengthening the Reporting of Observational Studies in Epidemiology reporting guideline was followed (online supplemental file). Eligible patients were identified from the database using the following International Classification of Diseases-10 codes: K550 (NDNL, GK7L and PR7K), K551 (F1A5, BPJQ and G440), and K559. IC diagnosis was confirmed based on clinical symptoms and laboratory findings, as internationally standardised diagnostic criteria for this condition are currently lacking. We used the diagnostic criteria described by Marston *et al*¹³ and are commonly used in Japan: (1) sudden abdominal pain with diarrhoea and bloody stools; (2) typical findings on contrast-enhanced CT (oedematous thickening of the intestinal wall and increased peri-intestinal fatty tissue concentration) or colonoscopy (ulceration/erosion, redness, oedema, biopsy image); (3) CT scan not suggestive of CRC at the onset of IC (asymmetric bowel wall thickening, polypoid intraluminal mass and exophytic mass); (4) no trauma or mechanical factors (eg, hernia);

(5) no evidence of severe acute ischaemic changes in other organs; and (6) patients who did not die due to the IC. Patients who did not undergo CT at the time of diagnosis were excluded. This study was approved by the ethical review board of Isesaki Municipal Hospital (approval number 2025–4). Consent from participants was obtained through an opt-out method on the institution's website.

Data extraction and definitions

We collected the following data from the patients' medical records: sex, age, performance status (Eastern Cooperative Oncology Group), previous history of colonoscopy, time from the last colonoscopy before IC onset, history of abdominal surgery, comorbidities, medications, symptoms at onset, extent of disease involvement of the IC, length of hospital stay, time from onset of IC to follow-up colonoscopy, number of patients with polyps on colonoscopy, number of polyps detected on colonoscopy, locations of polyps, macroscopic type of polyps, size of polyps and histopathological diagnoses. The extent of IC involvement was determined based on the endoscopic and CT findings. A history of colonoscopy was defined as having undergone a colonoscopy >90 days before the onset of IC to exclude the diagnosis of concomitant CRC with IC. Polyps were categorised based on macroscopic features into non-polypoid (0-IIa, IIb and IIc), pedunculated (0-Ip) and sessile (0-Is) types according to the Paris classification.¹⁴ Histological classification followed the WHO criteria.¹⁵ We defined endoscopic adenomas as endoscopically diagnosed adenomas (based on endoscopic findings using narrow-band imaging) when the histopathological diagnosis of the polyp could not be made (because the polyp was not resected).

Colonoscopic definitions and indications

A complete colonoscopy is defined as insertion of the colonoscope into the cecum or, in postcolectomy patients, to the anastomosis with the small intestine, regardless of bowel preparation quality. In contrast, an incomplete colonoscopy is defined as any colonoscopy where the whole colon is not observed. Colorectal neoplasms detected on incomplete colonoscopy were not double-counted even if they were subsequently detected on complete colonoscopy. The decision to perform a colonoscopy immediately after IC onset was left to the discretion of each treating physician. During colonoscopy immediately after the onset of IC, the extent of the involved area of IC only was observed, but a complete colonoscopy was performed if endoscope insertion was considered safe. Colonoscopy after IC recovery was performed according to the patient's preference, based on the physicians' recommendations.

Statistical analyses

Continuous variables are expressed as medians with ranges, and categorical variables are expressed as numbers with frequencies. Continuous variables were

Table 1 Baseline patient characteristics of the complete colonoscopy and incomplete/no colonoscopy groups

Variables	Complete colonoscopy	Incomplete/no colonoscopy	P value
Total number, n	116	280	
Sex (M/F), n (%)	36 (31%)/80 (69%)	65 (23.2%)/215 (76.8%)	0.1281
Age (years), median (range)	65 (21–88)	69 (17–97)	0.0229
Performance status 2, 3, 4	0	27 (9.6%)	0.0001
Previous history of colonoscopy, n (%)	24 (20.7%)	78 (27.9%)	0.1649
Time from the last colonoscopy before ischaemic colitis to ischaemic colitis onset (months), median (range), n (%)	35 (3–136)	31 (3–146)	0.83
History of abdominal surgery, n (%)	44 (37.9%)	108 (38.6%)	1
Comorbidities, n (%)			
Hypertension	47 (40.5%)	123 (44%)	0.5776
Hyperlipidaemia	24 (20.7%)	63 (22.5%)	0.7899
Diabetes	22 (19%)	41 (14.6%)	0.2932
Congestive heart failure	3 (2.6%)	7 (2.5%)	1
Coronary heart disease	4 (3.4%)	21 (7.5%)	0.1736
Chronic kidney disease	2 (1.7%)	11 (3.9%)	0.3613
Liver cirrhosis	0	3 (1%)	0.5589
Phleboscrotic colitis	0	0	1
Systemic lupus erythematosus	0	0	1
Medication, n (%)			
Diuretic drug	6 (5.2%)	6 (2.1%)	0.1181
Antibiotics	0	4 (1.4%)	0.3258
Digoxin	0	1 (0.3%)	1
NSAIDs	2 (1.7%)	18 (6.4%)	0.0743
Oral contraceptives	0	4 (1.4%)	0.3258
Anticancer drugs	1 (0.9%)	4 (1.4%)	1
Immunosuppressive drugs	0	0	1
Antiplatelet drugs	12 (10.3%)	34 (12.1%)	0.731
Anticoagulants	4 (3.4%)	7 (2.5%)	0.7375
Antiviral medicines for influenza	0	0	1
Saline laxatives	4 (3.4%)	28 (10%)	0.0407
Stimulant laxatives	1 (0.9%)	13 (4.6%)	0.0752

F, female; M, male; NSAID, non-steroidal anti-inflammatory.

compared using the Student's *t*-test, whereas categorical variables were compared using Fisher's exact test. Statistical significance was set at two-tailed $p < 0.05$. Statistical analyses were performed using the IBM SPSS Statistics for Windows (V.27.0; Tokyo, Japan).

RESULTS

Clinical features of IC and colonoscopy

The study flowchart is shown in online supplemental figure 1. A total of 418 patients were diagnosed with IC, of whom 22 were excluded because they did not undergo a CT scan. Of these, 45 patients underwent 51 colonoscopies during hospitalisation, including 35 incomplete and 16 complete colonoscopies. Incomplete colonoscopy

revealed two invasive carcinomas in two patients, one of whom developed stenosis, leading to IC. Six patients, including the above two patients, underwent incomplete colonoscopies followed by complete colonoscopies during hospitalisation. Ten patients underwent complete colonoscopies during hospitalisation. Of the patients who underwent incomplete colonoscopies during hospitalisation, 11 underwent complete colonoscopies after discharge. No lesions were detected in the 18 patients who underwent only incomplete colonoscopies. Of the 351 patients who did not undergo colonoscopies during hospitalisation, 259 did not undergo colonoscopies after discharge. Although 92 patients were scheduled for colonoscopy after discharge, three underwent incomplete

Table 2 Ischaemic colitis symptoms, locations and outcomes in the complete colonoscopy and incomplete/no colonoscopy groups

Variables	Complete colonoscopy	Incomplete/no colonoscopy	P value
Symptoms at onset, n (%)			
Abdominal pain	114 (98.2%)	259 (92.5%)	0.3613
Bloody stools	110 (94.8%)	268 (95.7%)	0.7914
Diarrhoea	114 (98.2%)	273 (97.5%)	1
Extent of disease involvement of ischaemic colitis, n (%)			
Cecum	0	2 (0.7%)	1
Ascending colon	1 (0.9%)	9 (3.2%)	0.2931
Transverse colon	20 (17.2%)	65 (23.2%)	0.2263
Descending colon	99 (85.3%)	264 (94.3%)	0.005
Sigmoid colon	57 (49.1%)	137 (48.9%)	1
Rectum	5 (4.3%)	23 (8.2%)	0.2004
Hospital length of stay (days), median (range)	5 (0–77)	5 (0–93)	0.485

colonoscopy because of inadequate bowel preparation or severe pain during the procedure, and three lesions were found. Complete colonoscopy was performed in 89 patients, and two invasive carcinomas were found. Therefore, complete colonoscopy was performed in 116 patients, incomplete colonoscopy in 21 and 259 did not undergo colonoscopy. No adverse events with the potential to be life-threatening, such as perforation or bleeding, were observed in any of the colonoscopies. [Table 1](#) presents the comparison of the baseline patient characteristics in the complete colonoscopy and incomplete/no colonoscopy groups. The patients in the incomplete/no colonoscopy group were significantly older, had a higher proportion of poor performance status and were more likely to have used saline laxatives. Complete colonoscopy was performed in 9.5% (7/74), 18.5% (5/27) and 35% (105/295) of patients aged >80, <40 and >40 but <80 years, respectively.

IC symptoms, locations and outcomes

[Table 2](#) presents the symptoms and locations of the IC in the complete and incomplete/no colonoscopy groups. Most patients exhibited symptoms involving abdominal pain, bloody stools and diarrhoea. Oedematous thickening of the intestinal wall on CT was localised in all patients, with the left side of the colon being the most frequently affected. The descending colon was more commonly affected in the incomplete/no colonoscopy group. In the colonoscopy and incomplete/no colonoscopy groups, 102/116 (87.9%) and 242/280 (86.4%) patients, respectively, required hospitalisation with a median stay of 5 days in both groups.

Table 3 Prevalence of colorectal neoplasms after ischaemic colitis following complete and incomplete colonoscopy

Variables	Incomplete colonoscopy	Complete colonoscopy
Total number, n	38	116
Time from onset of ischaemic colitis to colonoscopy (days), median (range)	1 (0–298)	48 (0–316)
Number of patients with polyps on colonoscopy, n (%)	5 (13.2%)	38 (32.8%)
Number of polyps detected on colonoscopy	5	73
Location of polyps, n (%)		
Cecum		5 (6.8%)
Ascending colon		16 (21.9%)
Transverse colon	2 (40%)	24 (32.9%)
Descending colon		4 (5.5%)
Sigmoid colon	3 (60%)	18 (24.7%)
Rectum		6 (8.2%)
Macroscopic type of polyps, n (%)		
Flat	1 (20%)	26 (34.7%)
Protruded	3 (60%)	40 (53.3%)
Pedunculated		7 (9.3%)
Type 2	1 (20%)	2 (2.7%)
Size of polyps (mm), median (range)	10 (7–65)	5 (2–25)
Pathology results		
Low-grade dysplasia	1 (20%)	20 (27.3%)
High-grade dysplasia		6 (8.2%)
Adenocarcinoma	2 (40%)	2 (2.7%)
Sessile serrated lesion		2 (2.7%)
Non-specific lesions (non-neoplastic)		1 (1.4%)
Endoscopic adenomas, n (%)	2 (40%)	42 (57.5%)

Polyps were categorised based on macroscopic features into flat (0-Ila, IIb and IIc), pedunculated (0-Ip), and protruded (0-Is) types according to the Paris classification, and type 2 according to the Bormann classification.

Prevalence of colorectal neoplasms after IC

[Table 3](#) presents the prevalence of colorectal neoplasms after IC, as determined using colonoscopy. Of the 38 patients with incomplete colonoscopies, two invasive adenocarcinomas were detected in 35 patients who were examined during hospitalisation, and three polyps were detected in three patients who were examined after discharge. Of the 116 patients who underwent complete colonoscopy, 73 polyps were detected, including two invasive adenocarcinomas. Additionally, 58% (42/73)

Table 4 Details of the four patients with colorectal cancer and ischaemic colitis

Patient number	Sex	Age, years	Previous history of colonoscopy	Colonoscopy	Location of ischaemic colitis	Location of tumour	Size, mm	Morphology	Pathology (T stage)
1	Female	74	Absent	Incomplete	S	S	65	Type 2	T3
2	Female	75	Absent	Incomplete	S, D	S	10	0-Is	T1
3	Male	62	Absent	Complete	S, D	T	12	0-Is	T1
4	Female	73	Absent	Complete	S, D	R	25	Type 2	T2

D, descending colon; R, rectum; S, sigmoid colon; T, transverse colon.

of lesions were ≤ 5 mm in diameter. Furthermore, in the complete colonoscopy group, including lesions that had been detected using incomplete colonoscopy, polyps were identified in 40 (34.5%) patients with 75 lesions, and invasive carcinoma was found in 4 (3.4%) patients. [Table 4](#) presents the details of the patients with CRC. Only one of the four patients was presumed to have developed IC due to stenosis caused by advanced cancer; one had early stage CRC at the IC site, which was presumed to be unrelated to the development of the IC, as no stenosis was observed; one had CRC in the proximal colon at the IC site; and one had CRC in the distal rectum. Patients 2 and 3 underwent endoscopic resection as the initial treatment, whereas patients 1 and 4 underwent surgery. High-grade dysplasia was detected in six cases according to the WHO criteria, five of which were adenomas and one of which was an intramucosal carcinoma according to the ninth edition of the Japanese Classification of Colorectal Carcinoma.^{16 17} [Table 5](#) presents the number of patients and polyp detection rates for each age group using complete colonoscopy. No significant difference in the polyp detection rates based on a history of colonoscopy was observed in any age group.

DISCUSSION

This study demonstrated the detection rates of colorectal polyps in patients with IC. The prevalence of CRC was 3.4% (4/116) and 4.3% (5/116) in the complete colonoscopy group, with the inclusion of intramucosal

cancer. Of the four patients with invasive CRC, two lesions were detected using complete colonoscopy, and one was located proximal to the involved IC site. In addition, 61% (45/73) of the polyps were located in the right-sided colon. Previous Japanese studies have reported a CRC detection rate of 0.73% during primary screening via colonoscopy and 3.9% during colonoscopy following a positive faecal immunochemical test.¹⁷ In addition, the detection rate of CRC in patients after colonic diverticulitis, in whom colonoscopy is recommended after onset by guidelines, as in IC, is reported to be 0%–1%.^{18–20} CRC is associated with diverse risk factors, including older age, obesity, diabetes mellitus, smoking, high consumption of red or processed meat and low physical activity.^{21 22} IC commonly occurs with older age, hypertension, diabetes mellitus, heart failure and the use of certain medications.^{7 10} Patients with IC may be at a high risk of CRC due to overlapping risk factors for CRC, such as older age, diabetes mellitus and smoking. Although the ACG guidelines recommend limiting endoscopic insertion to the distal extent of ischaemic involvement for the diagnosis of IC to prevent perforation or worsening of ischaemia, a substantial number of CRC cases and adenomas can be detected via complete colonoscopy in patients with IC.

None of the four patients with invasive CRC had a history of colonoscopy before the IC. Hence, prior colonoscopy history may be a factor in determining the need for colonoscopy after IC. Conversely, no difference was observed between colonoscopy history and polyp detection rates. This was presumably because 58% (42/73) of the detected lesions were ≤ 5 mm, and the Japanese guidelines until 2020 generally recommended against resecting polyps ≤ 5 mm, even when detected.^{22 23}

In this study, 29% of the patients diagnosed with IC after CT underwent complete colonoscopy. A selection bias may have occurred in the complete and incomplete/no colonoscopy groups. The incomplete/no colonoscopy group comprised patients with poor performance status and those older than 80 years and younger than 40 years. Younger patients may have had a lower CRC prevalence and, therefore, did not undergo intense colorectal examinations. Patients with a poor performance status were not considered for endoscopy, as they were unlikely to tolerate surgery if asymptomatic, and colonoscopy was not expected to impact their prognosis of survival.

Table 5 Number of patients and polyp detection rates for each age group using complete colonoscopy

PDR	Total (n=116)	Previous history of colonoscopy (–) (n=94)	Previous history of colonoscopy (+) (n=22)	P value
Age, years				
<40	1/4 (25%)	1/3 (33%)	0/1 (0%)	1
40–59	12/37 (32.4%)	10/33 (30.3%)	2/4 (50%)	0.5821
≥60	25/75 (33.3%)	19/58 (32.8%)	6/17 (35.3%)	1

Therefore, we focused on patients other than those with poor performance status. Older adults were more likely to have a poor performance status, which may have resulted in a complete colonoscopy not being performed. In addition, many patients had a history of saline laxative use because of constipation. Constipation could have led to poor bowel preparation or other problems, and patients and treating physicians may have avoided colonoscopy. Therefore, colonoscopy was often performed in patients aged 40–80 years who had a good performance status and could be treated for CRC.

At our institution, patients with typical symptoms of IC (abdominal pain, diarrhoea and bloody stools) and showing characteristic findings of IC on CT scans without suspicion of other causes of colitis are often diagnosed with IC without undergoing a colonoscopy. This approach may lead to a higher number of IC cases diagnosed without colonoscopy than with the strategy that relies on colonoscopic confirmation. Therefore, compared with the strategy that includes colonoscopy at the time of IC diagnosis, non-IC cases may have been misclassified as IC and true IC cases may have been missed in other patients. In daily practice, performing colonoscopy for all patients with colitis is challenging, along with performing two colonoscopies, including an immediate incomplete colonoscopy for IC diagnosis and a complete colonoscopy at a later stage. Prioritising complete colonoscopy at a later stage represents a reasonable strategy. Our results offer a valuable basis for deciding whether colonoscopy should be performed later in patients with IC.

This study also has some limitations. First, this was a single-centre retrospective study, and not all patients underwent colonoscopy, which may have introduced selection bias favouring patients whose physicians had a higher suspicion of CRC. In addition, background information, such as tobacco use, diet and exercise, which are risk factors for the development of CRC and IC, was not available. However, the reason for performing/not performing colonoscopy was based on a practical decision, and our results are clinically relevant. Future prospective studies are warranted to investigate the causal relationship between IC and CRC. Second, this study did not involve a control group, including healthy individuals or patients with positive faecal immunochemical test results. Third, we were unable to study the number of patients exhibiting clinical symptoms of IC in whom CRC was suspected based on CT. Consequently, our results should be interpreted with caution. However, the finding of invasive CRC in 3.4% of patients who underwent colonoscopy is a notable finding and warrants further investigation in future prospective studies with larger sample sizes.

In conclusion, for patients with IC without CRC detected using CT, subsequent complete colonoscopy may be recommended to rule out malignancy in patients without poor performance status.

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